Review 1

PROJ2999 Capstone Project Introduction

MENTAL HEALTH ANALYSIS USING MACHINE LEARNING MODELS

M.V.S.SATYA MRUDULA – 2023000597 G.RISHIKAR REDDY– 2023001672 D.VIVEK – 2023002218
T.S.L.PRIYANKA – 2023004539

Guide: Dr. M. Ramakrishna Murty,Professor

Department of Artificial Intelligence and Data Science
GITAM School of CSE

GITAM Deemed to be University, Visakhapatnam



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Project context

Depression, anxiety and stress can be difficult to recognise early. Assessment commonly relies on self-reported symptoms and professional evaluation.

Proposed framework

Analyse demographic, behavioural, lifestyle, academic/workplace, social and psychological factors for mental-health screening.

Intended output

An estimated risk or symptom category, with contributing factors explained. Target definitions depend on the selected dataset and validated assessment labels.

Scope: research prototype for screening and decision support. Clinical evaluation remains essential.

Introduction: Feature Research

Feature domain	Candidate variables from the research PDF
Demographic / history	Age, student or work status, family and personal history
Lifestyle / behaviour	Sleep quality and duration, exercise, days indoors, screen use
Academic / workplace	Pressure, study or work hours, satisfaction, performance
Social / financial	Peer pressure, support, loneliness, relationship and financial stress
Psychological	Coping struggles, mood changes, self-esteem, changes in habits

Proposed engineered features

Academic Pressure Index, Social Isolation Score, Digital Exposure Score and Support Gap require definitions and validation before use.

Exclude identifiers and target-derived predictors. Population prevalence data cannot label individual risk.

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Literature Review Overview

[1] ShamsEldin et al., 2025

Psychometric screening of depression, anxiety and stress.

[2] Nemesure et al., 2021

Biomedical and demographic predictors with SHAP explanations.

[3] Bailey and Strunk, 2023

Baseline predictors of cumulative symptoms over 12 months.

[4] Ahmed and Ahmed, 2023

Smartphone app usage for brief depression screening.

These four studies inform feature selection, model comparison and interpretation. Their outcomes and metrics differ.

Literature Review: Paper 1

ShamsEldin et al., Scientific Reports, 2025 [1]

Problem statement

Automated severity screening from questionnaire responses.

Main contributions

Compares five classifiers using DASS-42 and demographic information.

Inputs and outputs

In: 39,775 anonymised responses. Out: depression, anxiety and stress severity classes.

Methodology

1. DASS-42 + demographics
2. Clean and validate responses
3. Stratified split + fivefold CV
4. Compare five classifiers

Tools and technologies

SVM, Random Forest, Decision Tree, KNN and Naive Bayes.

Summary of results

SVM accuracy*	Value
Depression	99.3%
Anxiety	98.9%
Stress	98.8%

Major findings

SVM ranks highest. Questionnaire-derived targets need careful leakage checks and external validation.

Literature Review: Paper 2

Nemesure et al., Scientific Reports, 2021 [2]

Problem statement

Can routine health records flag depression and anxiety?

Main contributions

Combines model predictions and explains feature contributions using SHAP.

Inputs and outputs

In: biomedical and demographic records. Out: MDD and GAD predictions.

Methodology

1. Routine health records
2. Preprocess predictors
3. Train base models + stack
4. Evaluate held-out predictions

Tools and technologies

Stacked machine learning, XGBoost and SHAP.

Summary of results

Held-out ROC-AUC	Value
MDD	0.67
GAD	0.73

Major findings

Routine records contain useful signal, but discrimination is moderate. Generalisation needs further testing.

Literature Review: Paper 3

Bailey and Strunk, Journal of Clinical Psychology, 2023 [3]

Problem statement

Which baseline factors predict cumulative symptoms during the pandemic?

Main contributions

Selects predictors from 68 variables in 1,200 US adults.

Inputs and outputs

In: baseline psychosocial and demographic variables. Out: cumulative anxiety and depression severity.

Methodology

1. Eight assessments / 12 months
2. Impute + standardise
3. Elastic net with repeated CV
4. Test on held-out 30%

Tools and technologies

R 4.0.2, missForest, caret and glmnet.

Summary of results

Test R^2	Value
Anxiety	0.7818
Depression	0.7691

Major findings

Stress, sleep and psychological vulnerabilities inform feature selection. Pandemic context limits transferability.

Literature Review: Paper 4

M. S. Ahmed and N. Ahmed, JMIR Formative Research, 2023 [4]

Problem statement

Can brief access to phone usage support depression screening?

Main contributions

Mon Majhi retrieves seven days of app usage in about one second.

Inputs and outputs

In: usage features from 100 Bangladeshi students. Out: PHQ-9-based depression class

Methodology

1. Retrieve app usage history
2. Extract + select features
3. Leave-one-person-out CV
4. LightGBM prediction + SHAP

Tools and technologies

LightGBM, stable feature selection, Boruta and SHAP.

Summary of results

LightGBM metric	Value
Recall	82.4%
Precision	75.0%
F1-score	78.5%

Major findings

Phone behaviour offers screening signals. The small student sample needs broader validation.

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Summary of Literature Review

Ref.	Overview of paper	Model used	Novelty	Limitations
[1]	DASS-42 severity classification	SVM, RF, DT, KNN, NB	Common comparison across three outcomes	Self-report labels, conflicting metrics
[2]	Mental-health prediction from routine records	Stacked ensemble	Nonpsychiatric features with SHAP	Moderate discrimination, limited transferability
[3]	12-month cumulative symptom severity	Elastic net	Broad baseline feature selection	Pandemic context, follow-up attrition
[4]	App-usage depression screening	LightGBM, stacking	Fast retrospective data retrieval	100 students, questionnaire labels

Interpretation: The studies use different outcomes and evaluation metrics, so their reported performance is not directly comparable.

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Research Gaps

Synthesis of the four papers

The studies use different data sources, outcomes and populations. Their performance figures do not establish readiness for this project's target population.

Gaps addressed by the proposed project

1. Feature integration
Define a consistent, measurable feature set across the domains in the research PDF.

2. Reliable evaluation
Separate labels from predictors and test models under a shared validation protocol.

3. Interpretable screening
Evaluate explanations alongside sensitivity, calibration and subgroup performance.

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Problem Statement

Proposed research problem

Mental-health concerns may remain unrecognised until individuals report symptoms or receive professional evaluation.

This project will investigate whether machine-learning models can combine demographic, behavioural, lifestyle, academic/workplace, social and psychological factors to identify individuals who may be at risk of depression, anxiety or stress.

The framework will compare predictive models and explain contributing factors to support screening and professional decision-making.

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Project Objectives

Three objectives aligned with the research gaps

1. Establish a defensible feature framework
Consolidate the research PDF into a data dictionary. Define labels, measurement windows and exclusions for leakage or unavailable variables.

2. Compare predictive approaches
Evaluate Logistic Regression, Decision Tree, Random Forest, SVM, KNN and gradient boosting using consistent splits and screening metrics.

3. Develop interpretable screening outputs
Use SHAP or feature importance to explain predictions. Assess stability and subgroup performance, then demonstrate a screening prototype.

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Workplan: Proposed Methodology

1. Data and labels	Select an individual-level dataset and define screening outcomes.
2. Feature preparation	Audit missingness, encode variables and review candidate indices.
3. Model comparison	Fit preprocessing and feature selection inside training folds.
4. Evaluation	Use a held-out test set, threshold analysis and subgroup checks.
5. Interpretation and prototype	Explain risk estimates and show limitations in the screening output.

A shared feature catalogue does not imply that unrelated datasets can be merged by row.

Workplan and Proposed Timeline

Provisional 12-week plan from project approval

Work package	Owner	W1–2	W3–4	W5–6	W7–8	W9–10	W11–12
Data access, labels and scope	[Name]	Plan					
Feature dictionary and cleaning	[Name]	Plan	Plan				
Baseline model comparison	[Name]		Plan	Plan			
Tuning and held-out evaluation	[Name]			Plan	Plan		
Interpretation and subgroup checks	[Name]				Plan	Plan	
Prototype, report and review	[Team]					Plan	Plan

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Further Plan of Action

Confirm the study population

Choose the intended student or workplace setting and available outcome labels.

Audit the candidate datasets

Check permissions, provenance, missingness and whether the features coexist.

Finalise the feature dictionary

Define scales and time windows. Prioritise a concise, feasible input set.

Run reproducible baselines

Agree splits and metrics before tuning. Retain a simple comparison model.

Prepare the next review

Present the data audit, proposed architecture and initial results when available.

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1. T. ShamsEldin et al., “Artificial intelligence for predicting depression anxiety and stress using psychometric data,” *Scientific Reports*, vol. 15, Art. no. 37282, 2025. doi: 10.1038/s41598-025-21301-1.

2. M. D. Nemesure, M. V. Heinz, R. Huang, and N. C. Jacobson, “Predictive modeling of depression and anxiety using electronic health records and a novel machine learning approach with artificial intelligence,” *Scientific Reports*, vol. 11, Art. no. 1980, 2021. doi: 10.1038/s41598-021-81368-4.

3. B. Bailey and D. R. Strunk, “Predicting anxiety and depression over 12 months of the COVID-19 pandemic: A machine learning study,” *Journal of Clinical Psychology*, vol. 79, pp. 2388–2403, 2023. doi: 10.1002/jclp.23555.

4. M. S. Ahmed and N. Ahmed, “A Fast and Minimal System to Identify Depression Using Smartphones: Explainable Machine Learning–Based Approach,” *JMIR Formative Research*, vol. 7, e28848, 2023. doi: 10.2196/28848.

Feature-research source: Mental Health 2.pdf, supplied project notes, pp. 1–4.

Disclosure Statements

Academic integrity

All literature and source material must receive attribution. The team remains responsible for the submitted work and its claims.

AI assistance

AI assisted literature synthesis, drafting and slide preparation. Team review and compliance with institutional AI-use requirements remain necessary.

Data and research status

This deck presents a proposal. No original experimental results or clinical validation are claimed. Data access and ethical requirements must be confirmed.

Intended use

The proposed tool supports screening and professional evaluation. It does not provide a clinical diagnosis or replace treatment.